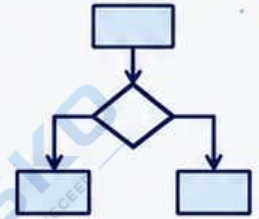


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Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



Data Science — Most Important Questions

Unit I - Introduction to Data Science

- **Answers in one or two lines (1 mark questions)**

1. What is Data Science?
2. State the importance of Data Science.
3. What is the Data Science lifecycle?
4. Name any two applications of Data Science.
5. What are the types of data?
6. What are Data Science tools and platforms?
7. What is Big Data?
8. What is Artificial Intelligence (AI)?

- **Short Answer Questions (3 marks questions)**

1. Explain Data Science and its importance.
2. Explain the Data Science lifecycle.
3. Explain applications of Data Science.
4. Explain different types of data.
5. Explain Data Science tools and platforms.
6. Write a short note on Big Data and AI.

- **Long Answer Questions (6 marks questions)**

1. Explain Data Science, its importance, lifecycle and applications.
2. Explain the Data Science lifecycle in detail.
3. Explain types of data, Data Science tools/platforms, Big Data and AI.

Unit II - Data Collection & Preprocessing

- **Answers in one or two lines (1 mark questions)**

1. What is data collection?
2. What is structured data?
3. What is unstructured data?
4. What is data cleaning?
5. What are missing values?
6. What is data transformation?
7. What is normalization?
8. What is data integration?
9. What is data reduction?
10. What is Python for Data Science?

- **Short Answer Questions (3 marks questions)**

1. Explain different data collection methods.
2. Differentiate between structured and unstructured data.
3. Explain data cleaning and handling missing values.
4. Explain data transformation and normalization.
5. Explain data integration and reduction.
6. Explain the role of Python in Data Science.

- **Long Answer Questions (6 marks questions)**

1. Explain data preprocessing and its major steps.
2. Explain data cleaning and handling missing values in detail.
3. Explain data transformation, normalization, integration and reduction.
4. Explain structured and unstructured data with suitable examples.

Unit III - Data Visualization & Statistics

- **Answers in one or two lines (1 mark questions)**

1. What is data visualization?
2. What is a chart?
3. What is a graph?
4. What is a Bar Chart?
5. What is a Pie Chart?
6. What is a Histogram?
7. What is a Scatter Plot?
8. What is descriptive statistics?
9. Define Mean.
10. Define Median.
11. Define Mode.
12. What is Correlation?
13. What is Regression?
14. What is probability?

- **Short Answer Questions (3 marks questions)**

1. Explain the importance of data visualization.
2. Explain Bar Chart, Pie Chart and Histogram.
3. Explain Scatter Plot with its uses.
4. Explain Mean, Median and Mode.
5. Explain Correlation and Regression basics.
6. Explain probability concepts in Data Science.

- **Long Answer Questions (6 marks questions)**

1. Explain data visualization and different charts/graphs with examples.
2. Explain descriptive statistics — Mean, Median and Mode — with examples.
3. Explain Correlation and Regression with suitable examples.
4. Explain probability concepts and their applications in Data Science.
5. Explain Bar Chart, Pie Chart, Histogram and Scatter Plot.

Unit IV - Introduction to Machine Learning

- **Answers in one or two lines (1 mark questions)**

1. What is Machine Learning?
2. What is supervised learning?
3. What is unsupervised learning?
4. What is classification?
5. What is regression?
6. What is clustering?
7. State any two applications of Machine Learning.
8. What are ethical issues in Data Science?

- **Short Answer Questions (3 marks questions)**

1. Explain Machine Learning and its basic concept.
2. Differentiate between supervised and unsupervised learning.
3. Explain classification and regression.
4. Explain clustering with an example.
5. Explain applications of Machine Learning in real life.
6. Explain ethical issues in Data Science.

- **Long Answer Questions (6 marks questions)**

1. Explain Machine Learning and its types — Supervised and Unsupervised Learning.
2. Explain Classification, Regression and Clustering.
3. Explain real-life applications of Machine Learning.
4. Explain ethical issues in Data Science.